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**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**

Order Instituting Rulemaking to Establish
Policies, Processes, and Rules to Ensure Reliable
Electric Service in California in the Event of an
Extreme Weather Event in 2021.

Rulemaking 20-11-003
(Filed November 19, 2020)

OPENING BRIEF OF THE VEHICLE-GRID INTEGRATION COUNCIL

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February 5, 2021

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission” or “CPUC”) and pursuant to the schedule set forth in the *Assigned Commissioner’s Coping Memo and Ruling* (“Scoping Memo”), issued by Commissioner Batjer on December 21, 2020, the Vehicle-Grid Integration Council (“VGIC”) ¹ hereby submits this opening brief addressing issues laid forth in the Scoping Memo.

1. INTRODUCTION.

VGIC is a 501(c)6 membership-based advocacy group committed to advancing the role of electric vehicles (“EV”) and vehicle-grid integration (“VGI”) through policy development, education, outreach, and research. VGIC supports the transition to a decarbonized transportation and electric sector by ensuring the value from EV deployments and flexible EV charging and

¹ VGIC member companies include American Honda Motor Co., Inc., Connect California LLC, Enel X North America, Inc., Fermata, LLC., Fiat Chrysler Automobiles, Ford Motor Company, General Motors Company, Nissan North America, Inc., Nuvve Corporation, The Mobility House, Toyota Motor North America, Inc., and Ossiaco, Inc. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all of the individual VGIC member companies or supporters. (<https://www.vgicouncil.org/>).

discharging is recognized and compensated in support of achieving a more reliable, affordable, and efficient electric grid.

2. ELECTRIC VEHICLES (THROUGH BOTH “V1G” AND “V2G” SOLUTIONS) CAN ALREADY CONTRIBUTE TO GRID RELIABILITY NEEDS IN 2021 AND 2022

Electric vehicles (“EVs”) are capable of both load reduction through managed charging (often referred to as “V1G”) or battery discharge to serve on-site load (often referred to as “V2L”) as well as serving as a temporary generation resource through exports of power to the grid (often referred to as “V2G”). Collectively, V1G, V2L, and V2G are often called Vehicle Grid Integration or VGI services. Until recently, VGI capabilities were considered to be a future technology. However, latent VGI capability already exists among EVs on the road today. This capability is not currently being fully utilized to support grid reliability during extreme weather events but could be leveraged to support grid needs in 2021 and 2022.

Existing V1G Capability

For example, an informal survey of VGIC member companies in December 2020 revealed that tens of thousands of vehicles in California already exist with the technical capability to receive a command to stop charging via telematics instructions from the original equipment manufacturer (“OEM”). Vehicle owner consent for this service could be provided through an app-based enrollment program. This service could readily be executed through a contracted payment to an OEM provider. Similarly, tens of thousands of charging stations already have the ability to schedule curtailment events (with pre-consent) due to alerts from an investor-owned utility (“IOU”) or the California Independent System Operator (“CAISO”). This service is available for Summer 2021. While VGIC acknowledges that

vehicle availability is a factor, the testimony of the California Energy Storage Association (“CESA”) points out that even a small fraction of the total vehicle population participating in these services could still amount to a meaningful contribution to grid reliability – potentially in the hundreds of MWs.²

Existing V2L/V2G Capability

By allowing discharge to serve on-site load, or export to the grid, vehicles with bi-directional V2L/V2G capability can effectively double an EV’s contribution to net peak load needs versus V1G or other forms of load-reduction. In recent months, steps have been taken by this Commission such that V2L/V2G is also now technically possible with EVs that are already on the road today. Most notably, through Decision 20-09-22, the Commission updated its Rule 21 interconnection standards to enable a permanent interconnection pathway for vehicles with a V2G-DC configuration (requiring an external device) and a temporary pathway for vehicles with a V2G-AC configuration (embedded in the vehicle itself). At this time, tens of thousands of vehicles with V2G-DC capability exist in California in the form of the Nissan Leaf and some electric school buses, if specifically-designed Electric Vehicle Supply Equipment (EVSE) is installed to provide the V2L/V2G functionality. UL-certified commercial EVSE products (such as those offered by VGIC member companies Fermata and Nuvve) are already available to enable this V2G functionality and could be deployed before the summer 2021 timeframe.

Despite these technical advances, certain administrative barriers still remain for V2L/V2G to effectively contribute to grid reliability in 2021. For example, Nuvve – a global V2G leader – was recently notified by Southern California Edison (“SCE”) that its Rule 21-

² CESA Direct at 29.

compliant bidirectional chargers do not qualify to participate in SCE's Charge Ready program and therefore cannot use the program's funds to offer V2L/V2G capabilities in chargers it is actively deploying. Fermata Energy, another leading provider of V2L/V2G services, has also found similar barriers.³ To briefly summarize, the administrative barriers include the following:

- ***CEC's smart inverter list:*** CEC's approved list of smart inverters does not currently V2L/V2G inverters, despite them being approved for interconnection.
- ***Utility interconnection approval:*** VGIC member companies have experienced difficulties receiving utility interconnection approval under a "continuous export" scheme due concerns over customer satisfaction.
- ***Utility Transportation Electrification (TE) program requirements:*** TE programs often require new service lines to be established (which can add time and cost to V2L/V2G projects being developed) when these projects could instead be deployed at existing services. Additionally, as mentioned above, utility TE programs that are currently being administered under SB 350 do not allow for funds to be applied to V2L/V2G devices.

Additionally, to VGIC's knowledge, there are no existing rates or programs offered by the IOUs that would appropriately compensate customers for V2G exports during net peak load hours. Existing transportation electrification programs also require a separate service, making baselining measurement difficult as required under a traditional demand response framework. VGIC believes that future Commission Decisions in this proceeding could play an instrumental role in overcoming some of these barriers, if not for summer 2021, then at least for summer

³ See Fermata's October 23, 2020 comments in SGIP OIR (R.20-05-012)

2022. VGIC notes that several more V1G and V2G capable vehicles are expected to be on the road in 2022 and future years that could also take advantage of such developments.

3. EV’S & EVSE’S SHOULD BE EXPLICITLY CONSIDERED AMONG THE DEMAND SIDE MEASURES THE COMMISSION DIRECTS THE IOUS TO PURSUE IN FUTURE DECISIONS IN THIS PROCEEDING, INCLUDING IN THE 2021 TIMEFRAME.

Several parties in this proceeding testified on the details of existing EV-eligible or EV-specific Demand Response programs currently offered by the IOUs and community choice aggregators (“CCAs”). For example, SCE noted that “EV customers are eligible to participate in multiple demand response (“DR”) programs or pilots such as the capacity bidding program (“CBP”), the demand response auction mechanism (“DRAM”), and SCE’s Charge Ready DR pilot (“CRDRP”).”⁴

Pacific Gas & Electric (“PG&E”) noted that “DR programs and any third-party contracts (e.g., DRAM) are technology agnostic and currently allow retail customers and third-party providers (supporting retail customers) to use any technology, including EV and charging stations, to provide load reduction.”⁵ Furthermore, they described a new demonstration pilot being conducted as part of their Demand Response Emerging Technology Program.

The California Community Choice Association (“CalCCA”) described multiple existing EV DR programs being pursued by the CCAs including those by Clean Power Alliance (“CPA”) (Power

⁴ SCE Direct at 43

⁵ PG&E at 4-21

Response), Silicon Valley Clean Energy (GridShift), Sonoma Clean Power (GridSavvy), and Pioneer Clean Energy (Flex Charging Pilot).

VGIC believes these existing EV DR programs should be leveraged to their full extent to reduce net peak load during critical hours in summer 2021. In the case of the IOUs this may require additional resources applied to support incremental procurement activities through these programs.

In addition, the Commission should consider directing the IOUs to conduct solicitations for new demand-side measures focused on EVs prior to summer 2021. These solicitations could be targeted toward both EVSEs with existing abilities to schedule curtailment events as well as OEMs with existing ability to stop charging via telematics (as described in Section 2 above).

Finally, EV's should be considered as a potential participating resource under ELRP efforts that are developed as part of this proceeding.

4. FUTURE DECISIONS IN THIS PROCEEDING SHOULD INCLUDE MEANINGFUL ACTIONS TO BE TAKEN THROUGH THE DRIVE OIR PROCEEDING (R. 18-12-006) TO SCALE UP V1G, V2L, AND V2G RESOURCES DURING NET PEAK LOAD HOURS BEFORE SUMMER 2022

While actions the Commission can take to increase EV contributions to summer reliability in 2021 are more limited, there are many things that could be done in the 2022 timeframe.

In particular, the DRIVE OIR proceeding (R.18-12-006) can and should be leveraged for this purpose. There are several areas where additional efforts in the DRIVE proceeding could be useful towards alleviating 2022 summer net load peak.

Demand Response Workshop

For example, PG&E's testimony notes that "a decision adopted in December 2020 calls for a joint IOU workshop to be held in the first quarter of 2021 to educate potential Vehicle Grid Integration (VGI) providers on DR opportunities and identify any barriers to participation for VGI resources."⁶

While the DR workshop PG&E mentions is a constructive step, the Decision referenced (D.20-12-029) fell short of requiring any meaningful follow-on action (besides a post-workshop report) that might lead to incremental VGI DR resources being procured prior to summer 2022. VGIC joined with nine other parties ("Joint Commenters") to this proceeding to point out this shortcoming.⁷ Future decisions in this proceeding should aim to resolve this issue.

Submetering Protocol

In parallel to the DR workshop and procurement activities mentioned above, a critical step for enabling DR from EVs is to resolve the outstanding issues regarding submetering protocols, which have been a persistent barrier to wider EV participation in DR programs. Future decisions in this proceeding should also aim to resolve this issue.

VGI Pilots

A major provision in the December 2020 VGI Decision (D.20-12-029) was the authorization of up to \$35 million in ratepayer-funded VGI Pilot programs what will be developed over the coming months. This could be a constructive avenue for targeted development of new VGI-enabled DR or load

⁶ PG&E at 4-22 referencing D.20-12-029, OP 9.

⁷ R.18-12-006, Joint Opening Comments Of The Vehicle-Grid Integration Council, American Honda Motor Co., Inc., Enel X North America, Inc., Advanced Energy Economy, California Energy Storage Alliance, Environmental Defense Fund, EVBox North America, Greenlots, Natural Resources Defense Council, And Siemens On Proposed Decision Concerning Implementation Of Senate Bill 676 at 6.

reduction activities prior to summer 2022. Future decisions in this proceeding could encourage these to-be-developed VGI Pilots to include a focus on summer reliability issues.

V2G Credit for Exports

Finally, D.20-12-029 fell short in terms of providing a concrete pathway for compensating V2G energy exports during net peak load hours. While it included supportive language on the general concept of credit-for-exports stating that “exploration of such a scheme should be pursued” there were no tangible steps identified regarding how such a scheme should be pursued, by when, and in which proceeding.

The Proposed Decision⁸ on this matter referenced the current Net Energy Metering (“NEM 3.0”) proceeding (R.20-08-020) as a potential venue to explore compensation for EV exports, however as the Joint Commenters pointed out, the Scoping Ruling issued in the NEM proceeding did not include consideration of exports from EVs.⁹ This fact was ignored in the final Decision (D.20-12-029) meaning that there is currently no venue for exploring broad-based compensation for V2G exports during net peak load hours that could in turn alleviate strained grid conditions during extreme weather events.

This oversight is especially concerning because broad-based adoption of V2G capabilities through a credit-for-export scheme could essentially double the resource adequacy (“RA”) capacity of EVs and EV Service Providers (“EVSPs”). Thus, it is an essential building block for unlocking a potentially valuable grid resource from EVs that can help avoid rotating outages during heat storms. From a long-term resource planning perspective, if California meets its EV deployment forecasts, then there will be the equivalent of thousands of MWs of batteries sitting in garages over the next few years that

⁸ R.18-12-006, Proposed Decision Concerning Implementation of Senate Bill 676 And Vehicle- Grid Integration Strategies the Joint Commenters.

⁹ See Joint Assigned Commissioner’s Scoping Memo and Administrative Law Judge Ruling Directing Comments on Proposed Guiding Principles issued in R.20-08-020 on November 19, 2020.

could be leveraged at relatively low cost to meet summer net peak load. This could in turn displace the costs of other more expensive generation resources -- especially in location constrained areas. However this will only happen if there is a clear incentive for exports to be compensated such that the V2G capability is embedded in the EV/EVSP equipment manufactured over the coming years.

Respectfully submitted,

A handwritten signature in black ink, appearing to read "Edward Burgess", with a stylized, cursive script.

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VEHICLE-GRID INTEGRATION COUNCIL

February 5, 2020